

BRADLEY D. SCHMIDT

SOFTWARE ENGINEERING STUDENT

778-586-8196 | bradleyschmidt04@gmail.com | [linkedin.com/in/bradleyschmidt04](https://www.linkedin.com/in/bradleyschmidt04) | github.com/Shupo4

SUMMARY

Software Engineering student drawn to intricate technical puzzles. Skilled in C++, Python, and JavaScript, with a project background spanning GPU-based ray tracing, mobile apps, and embedded systems. Combines a passion for hands-on exploration with engineering fundamentals from TRU to produce reliable, well-tested solutions.

EDUCATION

Thompson Rivers University

3rd Year Bachelor of Engineering in Software Engineering (GPA: 3.44)

Kamloops, BC, Canada

Expected Graduation Apr. 2027

SKILLS

Technical Skills

Programming Languages: C++, Python, RUST, JavaScript, SQL, VHDL, Matlab

Developer Tools: Linux, Git, VS Code, Vim, Android Studio, Oracle SQL Developer

AI and ML: Google Gemini AI, OpenAI API

Frameworks: React, React Native, Node.js

Platforms: Firebase, GitHub

Hardware: Arduino, Raspberry Pi, Esp8266, FPGA

Personal Skills

Original Solutions: Skilled at finding unconventional approaches to complex tasks

Methodical Reasoning: Proficient at breaking down problems into logical, manageable steps

Curious Mindset: Embraces new concepts quickly and enjoys exploring abstract ideas

Autonomous Workstyle: Works independently to craft fresh solutions beyond standard methods

PROJECTS

FPGA Calculator | VDHL, Vivado, ModelSim, Basys 3 FPGA

Nov. 2024 – Nov. 2024

- Used Basys 3 FPGA to make a four function calculator.
- Interfaced the FPGA with 4-7 segment display, and PMOD KYPD keypad
- Implemented addition, subtraction, multiplication and division using behavioral modeling in VHDL.
- Collaborated with one partner for CENG 3010 to deliver a fully functional FPGA-based calculator within a 1-week project window

Ray tracing Engine | Rust, C++

Feb. 2025 – Present

- Built from scratch using low-level math to calculate ray intersections.
- Modeled light bending around simulated black hole, incorporating gravitational lensing equations directly into ray intersection logic.
- Added diffuse and metallic reflection models, skybox, and adjustable parameters.
- Currently working on implementing Vulkan support and hardware level ray tracing.

AquaFlora Mobile App | React Native, Google Gemini AI, Python, Realm Database

Jul. 2023 – Jan. 2025

- Engineered a Mobile app for aquarium management, for users to track water parameters, reminders, and fish stocking.
- Led a 14-day testing phase with 12 users, uncovering and resolving 10+ issues before public launch.
- Integrated Google Gemini AI for fish/plant/disease image recognition, achieving 85% accuracy in trials and optimizing chatbot care tips via Gemini API.
- Automated data collection and validation for 650+ fish species via a Python scraper, reducing manual entry by 95%.
- Set up a monthly subscription model and Google Calendar integration, generating early recurring revenue and improving user retention.

URepair | C++, React, JavaScript

Oct. 2024 – Nov. 2024

- Developed a web based platform in a 3-person team, connecting contractors to job opportunities as part of a group project.
- Created backend (C++), and frontend interface (Node.js)
- Implemented optimal algorithms, improving search times by over 60%.

WORK EXPERIENCE

Maintenance

Service Canada

Mar. 2020 – Present

Kamloops, BC, Canada

- Maintained building cleanliness during peak COVID lockdown, demonstrating responsibility and attention to detail.
- Performed tasks such as carpet shampooing, pressure washing, and general upkeep to ensure a safe environment.

Concession Worker

Senor Froggy

Feb. 2023 – Feb. 2024

Kamloops, BC, Canada

- Supported the main cook in a high paced concession environment during peak hockey event hours.
- Worked efficiently as part of a tight knit team under pressure, enhancing teamwork and time management skills.